

Enterprise Architecture

Week 10



Course:

Introduction to Business Process Management

Lecturer:

Rosita, S.Pd. , M.Pd.

Prepared by:

Name: Aisyah Currents

NIM (Student id): 25091397108

Class: International

UNIVERSITAS NEGERI SURABAYA
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Questions:

1. Explain the definition of Enterprise Architecture (EA) and why EA is considered important in creating alignment between business processes and information technology in an organization!

Discuss the concept of EA as the blueprint of an organization, its functions for operational efficiency, business innovation, and the alignment of business strategies with IT.

Answer:

Enterprise Architecture (EA) is a strategic framework that defines the structure and operation of an organization through a comprehensive blueprint that aligns business goals with information technology (IT) systems. EA is considered important because it helps create harmony between business processes and IT, ensuring that technology investments support the overall mission and objectives of the organization. According to Spewak (1993), EA serves as a roadmap that guides organizational transformation by optimizing operational efficiency, encouraging innovation, and facilitating strategic alignment between business and IT. Through EA, organizations can reduce redundancy, improve data consistency, and strengthen decision-making processes.

2. Describe the four **main components (layers)** in Enterprise Architecture Planning (EAP) according to Spewak, and explain the role of each layer in the process of corporate architecture planning!

The layers are: Initiation Layer, Current Business Model & System, Architecture (Data, Application, Technology), and Implementation/Migration Plan.

Answer:

According to Steven H. Spewak, Enterprise Architecture Planning (EAP) consists of four main layers.

1. Initiation Layer

This layer focuses on defining the scope, objectives, and management commitment for EA projects. It establishes the foundation for governance and stakeholder involvement. By doing so, it creates a strong framework that guides subsequent architectural activities, ensuring that the enterprise architecture development proceeds in a structured, well-coordinated, and goal-oriented manner.

2. Current Business and System Model Layer

This layer analyzes the current state of the organization's business processes and information systems to identify gaps and inefficiencies. By mapping out the current state, organizations gain valuable insights into areas that require improvement, which then serve as a foundation for designing an optimized and future-oriented enterprise architecture.

3. Architecture Layer (Data, Application, and Technology)

This layer defines the target architecture for data management, applications, and IT infrastructure that will support the organization's goals. By clearly defining the data, application, and technology architecture, this layer provides a blueprint for building a cohesive, scalable, and future-ready enterprise system.

4. Implementation/Migration Plan Layer

This layer provides a roadmap for transitioning from the current architecture to the desired future state through phased implementation, risk management, and resource allocation. It includes strategies for risk management to anticipate and mitigate potential challenges, as well as plans for allocating resources such as budget, personnel, and technology. Through phased implementation, organizations can gradually adopt new systems and technologies without disrupting ongoing operations, ensuring a smooth transformation toward a more efficient and aligned enterprise architecture.

3. Analyze **the differences between Enterprise Architecture Planning (EAP) and Traditional Information Systems** in terms of focus, strategic approach, methodology, stakeholder involvement, as well as documentation and modeling!

Answer:

Enterprise Architecture Planning (EAP) differs significantly from traditional information systems approaches in several aspects. EAP emphasizes *strategic alignment* and a long-term vision, while traditional systems focus mainly on short-term operational needs. In terms of methodology, EAP uses structured frameworks, modeling techniques, and documentation to plan future IT architectures, whereas traditional approaches often rely on ad-hoc development without integration. From the perspective of stakeholder involvement, EAP requires active participation from business leaders, IT managers, and end-users, ensuring that every technology initiative supports business objectives. In contrast, traditional systems are usually managed solely by the IT department. Overall, EAP promotes a holistic and integrated approach to enterprise-wide system planning.

4. Mention and explain **five main reasons** why organizations need to implement *Enterprise Architecture*!

Focus on the relationship between organizational goals and information systems, cost efficiency, data integration, component reuse, and system development standards.

Answer:

There are five key reasons why organizations should implement Enterprise Architecture:

1. Alignment with Organizational Goals

EA ensures that IT strategies directly support the organization's mission and objectives. By establishing a clear connection between business processes and technological solutions, EA ensures that every IT initiative supports the organization's strategic direction. This alignment helps improve decision-making, prioritize projects that deliver the most value, and ensure that technology investments contribute directly to achieving organizational goals.

2. Cost Efficiency

By reducing redundancy in applications and infrastructure, EA minimizes operational costs. As a result, organizations can allocate their budgets more effectively, invest in

innovation, and maintain sustainable cost management without compromising performance or quality.

3. Data Integration

EA promotes seamless data exchange across departments, improving information consistency. Consequently, departments can work together more effectively, using reliable data to support business operations and strategic planning.

4. Component Reusability

Standardized frameworks allow for the reuse of system components, saving time and resources. Leading to faster deployment and reduced maintenance efforts.

5. System Development Standards

EA establishes guidelines for system design and implementation, enhancing quality and maintainability. Organizations can avoid technical fragmentation, simplify integration, and enhance long-term system sustainability, ultimately supporting continuous innovation and organizational growth.

5. Discuss **the business benefits** obtained by organizations from implementing *Enterprise Architecture Planning (EAP)*, and provide examples of its application in the context of educational institutions!

System integration between old and new systems, cost efficiency, strategic planning, and gradual system migration.

Answer:

The implementation of Enterprise Architecture Planning (EAP) provides numerous business benefits, including better integration between legacy and new systems, cost reduction, improved resource utilization, and strategic planning for future IT investments. EAP helps organizations achieve transparency in system architecture, reduces duplication of data and applications, and ensures that technological changes are implemented gradually through a migration plan. In the context of educational institutions, for example, EAP can streamline administrative processes, integrate academic and financial data systems, and enhance decision-making by providing accurate, real-time information. This leads to improved service delivery for students, faculty, and management.